

When the assistant says no

Routing a municipal RAG chatbot's refusals to a human — Gemeinde Schwyz, road traffic and parking department

Artifact: github.com/reigenmann-collab/RAGChatBotSZ-n8n-HITL Video: [\[RAG chatbot with n8n HITL caseworker\]](#)

PROBLEM

A municipal assistant that answers road-traffic and parking questions has to know when not to answer, and then has to do something with the question it declined. For a parallel course I built a German-language RAG assistant for Gemeinde Schwyz, grounded strictly in its published documents ([original demonstrator, deployed without any human hand-off](#)). It is deliberately cautious: it withholds an answer when its composite confidence falls below the operating threshold, when a fee question retrieves no fee schedule, and when the topic is a legal dispute — there it does not attempt an answer at all.

The refusal was honest. The sentence that followed it was not. The assistant told the citizen their inquiry “*was forwarded to the responsible caseworker*” (translated) — and nothing had been. There was no department end. A system that declines to answer and then drops the question is worse than one that never declines, because it looks like it worked.

SOLUTION

On an escalation the assistant now renders one button. It opens an n8n form with the question, the routing decision, the confidence value and a model-written case summary already filled in; the citizen adds name, e-mail and consent. n8n issues a case reference, mails the citizen a confirmation with a response deadline, and mails the caseworker the prepared case with Reply-To set to the citizen, so one click answers the right person. The app gained one function; retrieval, scoring, coverage and routing were not touched. Run end to end on a synthetic inquiry, both mails arrived (see appendix).

STREAMLIT APP | N8N OWNS EVERYTHING FROM HERE



STACK — AND WHAT I REJECTED

Claude Code	Built the prototype and this increment. A project skill plus a written progression log carry the standing decisions between sessions, so a cold session does not undo reasoning it cannot see. Its workflow passed every check short of running n8n, then failed on first contact — see Limits.
n8n	Owns the case from the click onward. Chosen over building intake into the app: a hand-off should be visibly separate from the logic that decided to hand off. Five nodes, one straight line, reviewable in a screenshot.
Gemini 3.1 Flash Lite	Writes the German answer, the grounding check that dominates the confidence score, and the case summary the caseworker receives. REJECTED the reasoning-model sibling: ~12.8 s per call against 2–4 s, and it refuses to let you switch the thinking budget off. Every candidate had to pass the same gate — catch a fabricated price and an invented validity period as ungrounded.
Link, not webhook	REJECTED the server-side POST. The link is followed by the citizen's browser, not the app's server, so a localhost n8n stays reachable when the app runs elsewhere — for the demo operator; a cloud POST never could. No shared secret, nothing in the request path to fail. Public use would need a hosted n8n.
SMTP, not Gmail OAuth	REJECTED OAuth. Self-hosted n8n would have needed my own Google Cloud app for the same two e-mails; an app password gave them with no extra setup.
Five nodes, not a case store	REJECTED the first design — Postgres, a case table, two extra Streamlit modules. It demonstrates the same judgment at a fraction of the build and does not fit a two-minute video.

LIMITS — WHERE IT BREAKS, AND WHAT A HUMAN STILL CHECKS

- **The hand-off is an invitation, not a guarantee.** If the citizen does not click, nothing reaches the department. A webhook would close that gap and cost me the reachability above. I took the trade knowingly.
- **No case register.** The caseworker's inbox is the queue — no status, no audit trail. This contradicts the parent project's own requirement for logged decisions with reviewer initials, so I state it as a scope cut rather than let it read as an oversight.
- **The 0.82 threshold is a placeholder, not a calibrated value.** The 50-query evaluation set exists and has not been run. Any escalation rate I quoted would be provisional, so I quote none.
- **A human checks everything on an escalated case.** The model writes the summary and the open points; the caseworker mail labels them as model-generated and says to verify. No generated answer reaches a citizen once the case is escalated.
- **I do not trust AI-written code unreviewed.** The agent's workflow failed on first import: a wrong response mode, and a form path n8n dropped silently so the link led nowhere. Its Code node also put citizen free text into HTML mail unescaped. None of it was visible in the diff; all of it surfaced only when the thing was run. I tested the fix in a throwaway container before touching my live instance.
- **Demo scope only.** n8n runs on localhost, and the e-mails are marked as a prototype — they are not communications from Gemeinde Schwyz. Synthetic inquiries only; no real citizen data.

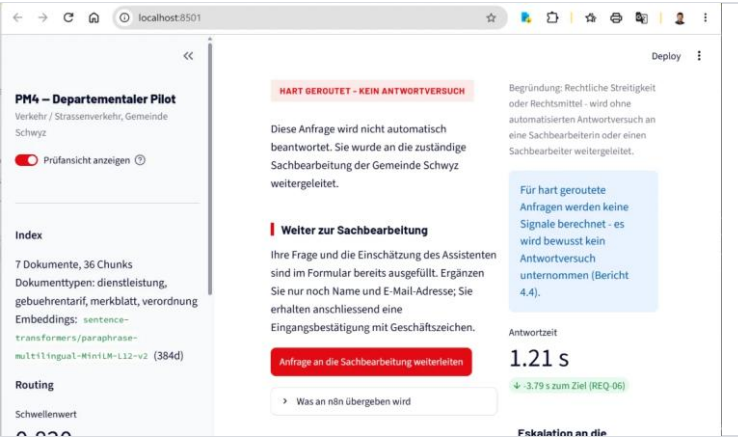
NEXT

Run the evaluation chain end to end and replace the placeholder threshold with a calibrated one. Until that exists I cannot say how often the assistant escalates — and that number, not the workflow, decides whether this is useful or merely safe. The harness is written and the test set is labelled against gold facts; what it needs is a caseworker to review 50 results. My estimate: about a day of work, most of it someone else's. The case register comes second, and only once there is evidence the volume justifies it.

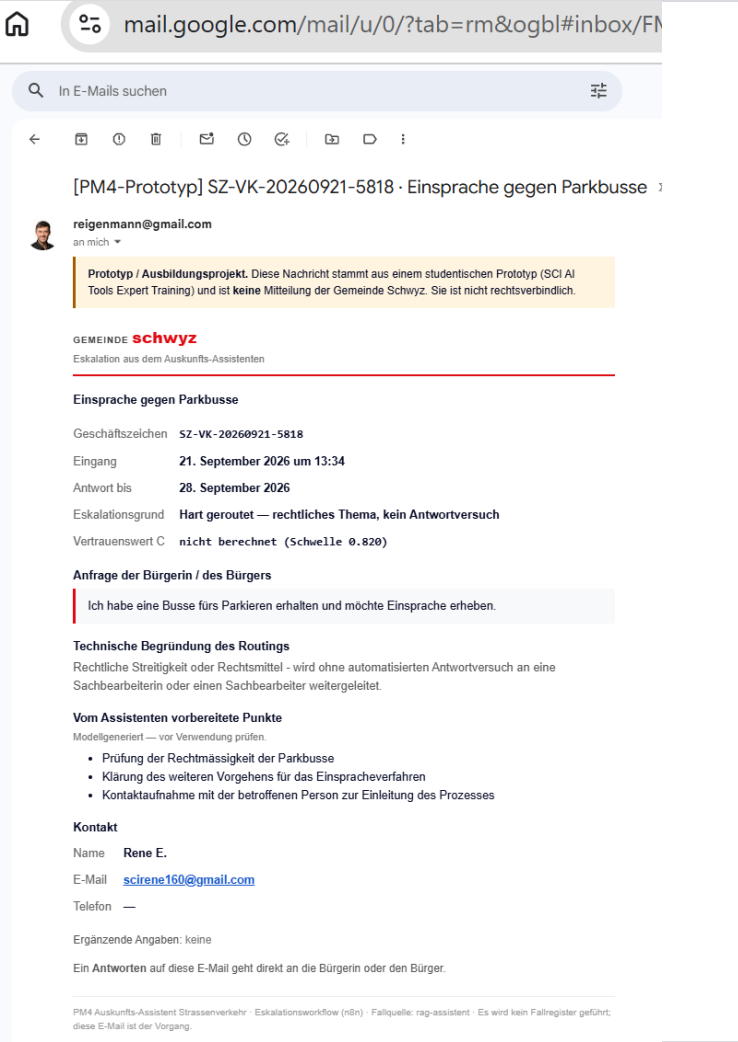
APPENDIX

The workflow running end to end

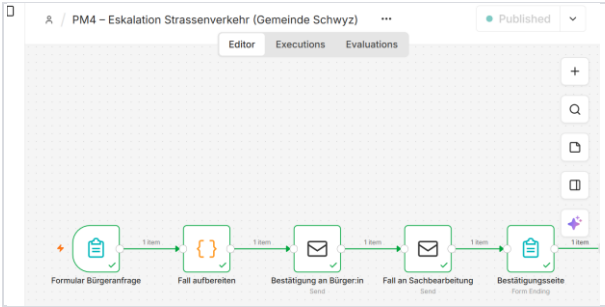
Screenshots from a local run on 21 September 2026, n8n 2.37 in Docker on localhost. Synthetic inquiries only.



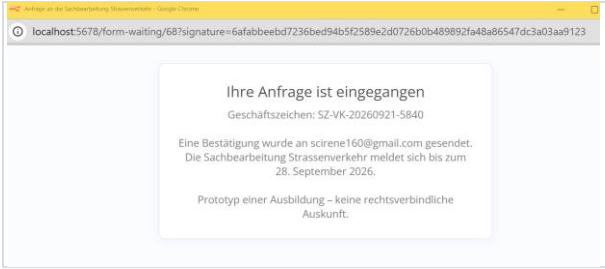
A The assistant's side. A hard-routed inquiry gets no answer attempt, then one button and an expander showing exactly what is handed to n8n. The app's part ends here.



B The caseworker's inbox. Case SZ-VK-20260921-5818, submitted through the assistant's link. A hard route, so the confidence reads "nicht berechnet". The model-written points are labelled "vor Verwendung prüfen"; Reply-To is the citizen.



C The n8n run. All five nodes executed; the workflow is published.



D What the citizen sees. Case reference and a five-working-day deadline, straight after submitting.



E The citizen's confirmation. Case SZ-VK-20260921-5818 mail submitted.